MUHAMMAD BAQER MOLLAH

University of Houston, TX 77479
• Email: mmollah@uh.edu • Web: https://mbaqer.github.io/

RESEARCH INTERESTS

• Security & Privacy • Deep Learning • Cyber-Physical Systems • Wireless Sensing & Communications

CURRENT POSITION

University of Houston September 2025 - Now
Dept. of Information Science Technology
Post-Doctoral Fellow
Advisor: Prof. Kyu In Lee
Lab: Smart Systems Security Laboratory (SSS Lab)
Project: USDOT Tier 1 UTC Transportation Cybersecurity Center for Advanced Research and Education (CYBER-CARE)
◦ Working closely with students to define research directions, plan project milestones, and oversee implementation and paper writing
◦ Designing and evaluating practical end-to-end security-aware mechanisms for connected vehicles

EDUCATION

University of Massachusetts Dartmouth Fall 2022 - Summer 2025
PhD in Electrical and Computer Engineering
Advisor: Prof. Honggang Wang, IEEE Fellow (Currently at Yeshiva University, NY)
Dissertation: On Enabling Multi-Modal Sensing and Security Techniques for Connected Vehicles
Courses Taken (Selected): Network Security, Multimedia Communications, Advanced Computer Systems, and Dependable & Secure Computing

Jahangirnagar University, Dhaka, Bangladesh 2015 - 2016
MS in Computer Science
Research Project: Secure Data Sharing and Searching at the Edge of Cloud Connected Smart Devices
Courses Taken (Selected): Artificial Neural Networks, Embedded System Engineering, Computer Networks, Mobile & Wireless Communications, Information Theory & Coding System, Distributed Systems, and Cloud Computing

International Islamic University Chittagong, Chittagong, Bangladesh 2009 - 2014
BS in Electrical and Electronic Engineering
Final Year Project: E-Police System for Improved E-Government Services in Bangladesh
Courses Taken (Selected): Electrical Circuits and Electronics, Digital Electronics, Computer Programming, Communication Theory, Numerical Analysis, Signals and Systems, Digital Signal Processing, and Control System

ACADEMIC EXPERIENCES

University of Massachusetts Dartmouth September 2022 - August 2025
Dept. of Electrical and Computer Engineering
Teaching Assistant & Research Assistant
◦ Assisted in teaching *Introduction to Engineering & Computing (Lab)* with basic programming related projects with C++ in Arduino, guided them in using introductory engineering tools, and provided one-on-one mentoring whenever necessary to address diverse learning needs

Last updated: August 13, 2026

◦ Helped manage lab sessions, monitored student progress, facilitated group discussions, and provided constructive feedback to support concepts
◦ Actively provided mentorship and guidance to the student teams to the semester-end group projects
◦ Conducted research in wireless sensing and communications for connected vehicles under the NSF funded project "Enabling Machine Learning based Cooperative Perception with mmWave Communication for Autonomous Vehicle Safety"

PUBLICATIONS

Google Scholar Profile: https://scholar.google.com/citations?user=VijQVZYAAAAJ&hl=en

* indicates mentored/co-mentored students

Submitted

[6] **M.B. Mollah**, K.I. Lee, "Toward Secure Internet of Things through Access Control Encryption", *IEEE Consumer Electronics Magazine*, pp. 1 -13, 2026. (Submitted)

[5] **M.B. Mollah**, K.I. Lee, "Ensuring In-Vehicle Network Security with Edge Computing and Access Control Encryption", *IEEE Transactions on Consumer Electronics*, pp. 1 -13, 2026. (Submitted)

[4] **M.B. Mollah**, K.I. Lee, "Practical End-to-End Security-Aware Mechanism for Connected Vehicles by Multi-Level Access Control", *ACM Transactions on Internet Technology*, pp.1-14, 2026. (Submitted)

[3] **M.B. Mollah**, K.I. Lee, "Multi-Level Secure Dissemination of Cooperative Awareness Messages in Internet of Vehicles," *2026 IEEE Future Networks World Forum (IEEE FNWF)*, Lisbon, Portugal, pp. 1-7. (Submitted)

[2] M.R. Shaik*, **M.B. Mollah**, Y. Zhang, L. Gao, K.I. Lee, "EV Charging Network Security by Enhanced PKI with Decentralized Certificates Revocation", *2026 IEEE Future Networks World Forum (IEEE FNWF)*, Lisbon, Portugal, pp. 1-6. (Submitted)

[1] M.R. Shaik*, **M.B. Mollah**, Y. Zhang, L. Gao, K.I. Lee, "Secure and Efficient Multicasting of Safety Messages for Connected Vehicles", *2026 IEEE Future Networks World Forum (IEEE FNWF)*, Lisbon, Portugal, pp. 1-7. (Submitted)

Journals/Magazines

[J5] **M.B. Mollah**, H. Wang, M.A. Karim, H. Fang, "Multi-Modal Sensing and Fusion in mmWave Beamforming for Connected Vehicles: A Transformer Based Framework", *IEEE Transactions on Vehicular Technology*, pp. 1-13, 2026.

[J4] **M.B. Mollah**, H. Wang, M.A. Karim, H. Fang, "Multi-Modality Sensing in mmWave Beamforming for Connected Vehicles Using Deep Learning", *IEEE Transactions on Cognitive Communications and Networking*, vol. 12, pp. 327-341, 2026.

[M2] **M.B. Mollah**, H. Wang, M.A. Karim, H. Fang, "mmWave Enabled Connected Autonomous Vehicles: A Use Case with V2V Cooperative Perception", *IEEE Network*, vol. 38, no. 6, pp. 485-492, Nov. 2024.

[J3] **M.B. Mollah**, M.A.K. Azad, Y. Zhang, "Secure Targeted Message Dissemination in IoT Using Blockchain Enabled Edge Computing", *IEEE Transactions on Consumer Electronics*, vol. 70, no. 3, pp. 5389-5400, Aug. 2024.

[J2] **M. B. Mollah**, J. Zhao, D. Niyato, Y. L. Guan, C. Yuen, S. Sun, K.-Y. Lam, and L. H. Koh, "Blockchain for the Internet of Vehicles towards Intelligent Transportation Systems: A Survey", *IEEE Internet of Things Journal*, vol. 8, no. 6, pp. 4157-4185, March 2021.

[J1] **M. B. Mollah**, J. Zhao, D. Niyato, K.-Y. Lam, X. Zhang, A. M.Y.M. Ghias, L. H. Koh, and L. Yang, "Blockchain for Future Smart Grid: A Comprehensive Survey", *IEEE Internet of Things Journal*, vol. 8, no. 1, pp. 18-43, January 2021.

[M1] **M.B. Mollah**, M.A.K. Azad, A. Vasilakos, "Secure Data Sharing and Searching at the Edge of Cloud-Assisted Internet of Things", *IEEE Cloud Computing*, Vol. 4, No. 1, January-February 2017, pp. 34-42.

Conferences

[C5] M. Adam*, **M.B. Mollah**, K.I. Lee, "Mapping Extended Reality Threats in Connected Vehicles with Attack-Defense Tree", *2026 IEEE 15th International Conference on Consumer Electronics - Berlin (ICCE-Berlin)*, Berlin, Germany, 2026, pp. 1-6. (Accepted)

[C4] **M.B. Mollah**, G. Zouridakis, K.I. Lee, "Securing the Lifeline: Enforcing Read and Write Access Control in Vehicular Safety Services", *IEEE International Conference on Cyber Security and Resilience (IEEE CSR)*, Lisbon, Portugal, 2026, pp. 1-6. (Accepted and Presented)

[C3] **M.B. Mollah**, H. Wang, H. Fang, "Evaluating Vulnerabilities of Connected Vehicles Under Cyber Attacks by Attack-Defense Tree", *IEEE International Conference on Computing, Networking and Communication (ICNC)*, Maui, Hawaii, USA, 2026, pp. 1-6.

[C2] **M.B. Mollah**, H. Wang, H. Fang, "Multi-Modal Sensing Aided mmWave Beamforming for V2V Communications with Transformers", *IEEE Global Communications Conference (GLOBECOM)*, Taipei, Taiwan, 2025, pp. 1-6.

[C1] **M.B. Mollah**, H. Wang, H. Fang, "Position Aware 60 GHz mmWave Beamforming for V2V Communications Utilizing Deep Learning", *IEEE International Conference on Communications (ICC)*, Denver, CO, USA, 2024, pp. 4711-4716.

Book Chapter

[B1] **M.B. Mollah**, S. Zeadally, M.A.K. Azad, "*Emerging wireless technologies for Internet of Things applications: Opportunities and challenges*", In Encyclopedia of Wireless Networks, Springer, 2020, pp. 390-400, Editors: Prof. Xuemin Shen, Prof. Xiaodong Lin, and Prof. Kuan Zhang.

STUDENT MENTORING

Muzamil Raheman Shaik, Master's Student, University of Houston — April 2026 – Present
Research Topic: Secure Multicasting for Connected Vehicles (Submitted to IEEE FNWF 2026)

Adam Mojtaba, Undergraduate Student, University of Houston — January 2026 – Present
Research Topic: Threat Modeling and Security Evaluation for Extended Reality Systems (Accepted to IEEE ICCE-Berlin 2026)

PROPOSAL WRITING EXPERIENCE

◦ **NSF - Collaborative Research:** CIF: Medium: Towards Fully Deep Learning-based mmWave Communications for Complex Mobile RF Environments, **PIs and Co-PIs:** Prof. Honngang Wang, Prof. Hua Fang, Prof. Hamid Sharif, UNL, **Status:** Not funded, **My Contribution:** One research thrust out of three in total.

RELATED TRAINING

Cybersecurity & Career Readiness — April 2026 to May 2026
Stevens Institute of Technology, NJ, USA
Supported by: National Institute of Standards and Technology (NIST).

AraFest Summer 2024 and 2025
ARA (Agriculture and Rural Communities) Annual Community Event
Iowa State University, Ames, Iowa, USA
Project: NSF Platforms for Advanced Wireless Research (PAWR) program.

SKILLS

Linux/Unix based operating systems (e.g., Ubuntu, Kali), Embedded systems (e.g., Raspberry Pi, Arduino), Networking tools, Virtualization, Ethical hacking, Latex, Python, MATLAB, Deep learning, PyTorce etc.

SERVICES

Web of Science Public Profile: https://www.webofscience.com/wos/author/record/T-4705-2019

Chair: Special Session on Cyber Resilience for Trustworthy Connected Consumer Mobility at IEEE 45th International Conference on Consumer Electronics (IEEE ICCE 2027), 8-11 January 2027, Las Vegas, NV, USA.

Reviewer for Journals/Magazines: IEEE Communications Magazine, IEEE Network, IEEE Wireless Communications Magazine, IEEE Transactions on Wireless Communications, IEEE Wireless Communications Letters, IEEE Journal on Selected Areas in Communications, IEEE Transactions on Mobile Computing, IEEE Transactions on Industrial Informatics, IEEE Transactions on Intelligent Transportation Systems, IEEE Transactions on Information Forensics and Security, IEEE Internet of Things Journal, IEEE Transactions on Network and Service Management, ACM Transactions on Computing for Healthcare, ACM Computing Surveys, Future Generation Computer Systems (Elsevier).

Technical Program Committee (TCP) Member: • Workshop on 5th Distributed and Integrated Communication, Sensing, and Computing in Space-Air-Ground Integrated Networks for 6G in ICCCN 2026, Hawaii, USA, • IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (IEEE SmartGridComm 2025), Toronto, Canada, • Workshop on Federated Learning for Autonomous and Connected Vehicles, 2026 IEEE 104th Vehicular Technology Conference (VTC2026-Fall), Boston, MA, USA, among others.

EasyChair Webmaster: The 33rd International Conference on Computer Communications and Networks (ICCCN 2024), July 29 - 31, 2024, Big Island, Hawaii, USA.

MEMBERSHIP

Member of • IEEE • IEEE Communications Society • IEEE Vehicular Technology Society

WORK AUTHORIZATION

US Permanent Resident (green card holder), no employer sponsorship required.

REFERENCES

Available, upon request.